

AthenaK EOS Implementation Note

Local code summary

May 6, 2026

1 Scope and Maintained Products

This note documents the maintained EOS table generators and dense-EOS helper modules in the current tree:

- `scripts/generate_saha_table.py`,
- `scripts/generate_lte_table.py`,
- `scripts/scvh95_eos.py`,
- `scripts/chabrier2021_eos.py`.

The public H/He LTE models exposed by the table generator are

1. `t13`,
2. `scvh_t13_cp_helm_union`,
3. `chabrier2021_t13_helm_union`.

The separate pure-H Saha table is intentionally kept as an independent, analytic reference EOS. Older experimental dispatch modes have been removed from the generator and from this note; private helper routines remain only where a maintained union still needs them.

2 Native Table Layout and Thermodynamic Closure

AthenaK-native EOS tables are sampled on natural-log axes

$$x \equiv \ln \rho_{\text{cgs}}, \quad y \equiv \ln T_{\text{K}}, \quad (1)$$

with temperature as the fastest-varying table index and `log_axis_base = e`. The runtime fields are

$$\ln P, \quad \ln \epsilon, \quad \ln c_s^2, \quad \Gamma_1, \quad \Gamma_3 - 1, \quad (2)$$

plus model-dependent composition and diagnostic fields.

For the finite-difference closures used by T13 and by dense fallback products, the code first builds completed surfaces $P(\rho, T)$ and $\epsilon_{\text{phys}}(\rho, T)$ and then evaluates

$$\chi_\rho = \left(\frac{\partial \ln P}{\partial \ln \rho} \right)_T, \quad \chi_T = \left(\frac{\partial \ln P}{\partial \ln T} \right)_\rho, \quad (3)$$

$$c_v = \frac{1}{T} \left(\frac{\partial \epsilon_{\text{phys}}}{\partial \ln T} \right)_\rho, \quad \Gamma_3 - 1 = \frac{P \chi_T}{\rho T c_v}, \quad (4)$$

$$\Gamma_1 = \chi_\rho + \chi_T(\Gamma_3 - 1), \quad c_s^2 = \Gamma_1 \frac{P}{\rho}. \quad (5)$$

The exported auxiliary heat capacity and adiabatic gradient are

$$c_p = c_v \frac{\Gamma_1}{\chi_\rho}, \quad \nabla_{\text{ad}} = \frac{\Gamma_3 - 1}{\Gamma_1}. \quad (6)$$

At the two ends of a grid direction the finite differences use second-order one-sided stencils; in the interior they use centered differences.

When photon radiation is enabled, it is added as an independent blackbody component,

$$P_{\text{rad}} = \frac{a_{\text{rad}} T^4}{3}, \quad \epsilon_{\text{rad}} = \frac{a_{\text{rad}} T^4}{\rho}, \quad s_{\text{rad}} = \frac{4a_{\text{rad}} T^3}{3\rho}, \quad (7)$$

and the diagnostic radiation pressure fraction is

$$\beta_{\text{rad}} = \frac{P_{\text{rad}}}{P_{\text{gas}} + P_{\text{rad}}}. \quad (8)$$

3 Pure-H Saha Reference EOS

`generate_saha_table.py` solves

$$\text{H} \rightleftharpoons \text{H}^+ + e^- \quad (9)$$

for a ground-state pure-H gas. The partition convention cancels the electron spin degeneracy against the neutral-H ground-state degeneracy, giving

$$C_{\text{Saha}} = m_{\text{H}} \left(\frac{2\pi m_e k_{\text{B}}}{h^2} \right)^{3/2}, \quad y = \frac{C_{\text{Saha}} T^{3/2} \exp(-T_{\text{ion}}/T)}{\rho}, \quad (10)$$

where $T_{\text{ion}} = \chi_{\text{H}}/k_{\text{B}}$. The ionization fraction $x = n_{\text{H}^+}/(n_{\text{H}} + n_{\text{H}^+})$ satisfies

$$\frac{x^2}{1-x} = y, \quad x = \frac{2y}{\sqrt{y^2 + 4y} + y}, \quad (11)$$

with asymptotic low- and high- y branches for numerical stability.

The pressure and specific internal energy are

$$P = \rho R_{\text{H}} T (1 + x), \quad \epsilon = \frac{3}{2} R_{\text{H}} T (1 + x) + x \frac{\chi_{\text{H}}}{m_{\text{H}}}, \quad R_{\text{H}} = \frac{k_{\text{B}}}{m_{\text{H}}}. \quad (12)$$

The table closure is analytic. Defining

$$g = \frac{x(1-x)}{2-x}, \quad u_{\text{H}} = \frac{T_{\text{ion}}}{T}, \quad (13)$$

the derivatives used by the generator are

$$\left(\frac{\partial \epsilon}{\partial \rho}\right)_T = -\frac{g}{\rho} \left(\frac{3}{2} R_H T + \frac{\chi_H}{m_H}\right), \quad (14)$$

$$\left(\frac{\partial \epsilon}{\partial T}\right)_\rho = \frac{3}{2} R_H (1+x) + \left(\frac{3}{2} R_H T + \frac{\chi_H}{m_H}\right) \frac{g(3/2 + u_H)}{T}, \quad (15)$$

$$\left(\frac{\partial P}{\partial \rho}\right)_T = R_H T [(1+x) - g], \quad (16)$$

$$\left(\frac{\partial P}{\partial T}\right)_\rho = \rho R_H [(1+x) + g(3/2 + u_H)]. \quad (17)$$

The adiabatic temperature derivative follows the first law,

$$\left(\frac{\partial T}{\partial \rho}\right)_s = \frac{P/\rho^2 - (\partial \epsilon / \partial \rho)_T}{(\partial \epsilon / \partial T)_\rho}, \quad (18)$$

and the stored sound speed and adiabatic exponents are

$$c_s^2 = \left(\frac{\partial P}{\partial \rho}\right)_T + \left(\frac{\partial P}{\partial T}\right)_\rho \left(\frac{\partial T}{\partial \rho}\right)_s, \quad \Gamma_1 = \frac{\rho c_s^2}{P}, \quad \Gamma_3 - 1 = \frac{\rho}{T} \left(\frac{\partial T}{\partial \rho}\right)_s. \quad (19)$$

This H-only EOS is not merged into the H/He union tables; its generated table is kept byte-for-byte stable when the LTE H/He products are updated.

4 T13 H/He Chemical-Equilibrium EOS

t13 implements the Tomida et al. Appendix-1 ideal chemical-equilibrium H/He EOS with molecular hydrogen. The species order is

$$\text{H}_2, \text{H}, \text{H}^+, \text{He}, \text{He}^+, \text{He}^{2+}, e^-. \quad (20)$$

The translational single-particle partition function is stored as

$$\ln z_{\text{tr}}(m, T) = \frac{3}{2} \ln(2\pi m k_B T) - 3 \ln h. \quad (21)$$

For H_2 , the rotational partition function uses a fixed 3:1 ortho/para ratio. With

$$Z_{\text{even}} = \sum_{J \text{ even}} (2J+1) \exp\left[-\frac{J(J+1)\theta_{\text{rot}}}{2T}\right], \quad (22)$$

$$Z_{\text{odd,shift}} = \sum_{J \text{ odd}} 3(2J+1) \exp\left[-\left(\frac{J(J+1)}{2} - 1\right) \frac{\theta_{\text{rot}}}{T}\right], \quad (23)$$

the code uses

$$\ln Z_{\text{rot}} = \frac{1}{4} \ln Z_{\text{even}} + \frac{3}{4} \ln Z_{\text{odd,shift}}, \quad (24)$$

with the same weights for $\partial \ln Z_{\text{rot}} / \partial \ln T$. The vibrational partition function is

$$a = \frac{\theta_{\text{vib}}}{2T}, \quad \ln Z_{\text{vib}} = -a - \ln(1 - \exp[-2a]), \quad \frac{\partial \ln Z_{\text{vib}}}{\partial \ln T} = \frac{a}{\tanh a}. \quad (25)$$

At large a the generator uses the limiting branch $\ln Z_{\text{vib}} \simeq -a$.

The implemented species partition offsets are

$$\ln z_{\text{H}_2} = \ln z_{\text{tr}}(m_{\text{H}_2}, T) + \ln Z_{\text{rot}} + \ln Z_{\text{vib}} + \ln 4 + \ln 2, \quad (26)$$

$$\ln z_{\text{H}} = \ln z_{\text{tr}}(m_{\text{H}}, T) + \ln 2 + \ln 2 - \frac{\chi_{\text{dis}}}{2k_{\text{B}}T}, \quad (27)$$

$$\ln z_{\text{H}^+} = \ln z_{\text{tr}}(m_{\text{H}}, T) + \ln 2 + \ln 2 - \frac{\chi_{\text{dis}} + 2\chi_{\text{ion,H}}}{2k_{\text{B}}T}, \quad (28)$$

$$\ln z_{\text{He}} = \ln z_{\text{tr}}(m_{\text{He}}, T), \quad (29)$$

$$\ln z_{\text{He}^+} = \ln z_{\text{tr}}(m_{\text{He}}, T) - \frac{\chi_{\text{ion,He1}}}{k_{\text{B}}T}, \quad (30)$$

$$\ln z_{\text{He}^{2+}} = \ln z_{\text{tr}}(m_{\text{He}}, T) - \frac{\chi_{\text{ion,He1}} + \chi_{\text{ion,He2}}}{k_{\text{B}}T}, \quad (31)$$

$$\ln z_e = \ln z_{\text{tr}}(m_e, T) + \ln 2. \quad (32)$$

The equilibrium constants are then

$$\ln K_{\text{dis}} = 2 \ln z_{\text{H}} - \ln z_{\text{H}_2}, \quad \ln K_{\text{ion,H}} = \ln z_{\text{H}^+} + \ln z_e - \ln z_{\text{H}}, \quad (33)$$

with analogous first and second helium ionization constants.

For each density row the generator solves the electron density by charge neutrality. The hydrogen atomic density is written in a stable quadratic form,

$$n_{\text{H}} = \frac{2n_{\text{H,tot}}n_e}{\sqrt{(n_e + K_{\text{ion,H}})^2 + 8n_{\text{H,tot}}n_e^2/K_{\text{dis}}} + n_e + K_{\text{ion,H}}}, \quad (34)$$

followed by

$$n_{\text{H}^+} = n_{\text{H}} \frac{K_{\text{ion,H}}}{n_e}, \quad n_{\text{H}_2} = \frac{n_{\text{H}}^2}{K_{\text{dis}}}. \quad (35)$$

For helium, the normalized weights are

$$w_0 = 1, \quad w_1 = \frac{K_{\text{He1}}}{n_e}, \quad w_2 = \frac{K_{\text{He1}}K_{\text{He2}}}{n_e^2}, \quad (36)$$

so that $n_{\text{He}^q} = n_{\text{He,tot}}w_q/(w_0 + w_1 + w_2)$. The bisection root is

$$n_{\text{H}^+} + n_{\text{He}^+} + 2n_{\text{He}^{2+}} - n_e = 0. \quad (37)$$

The ideal partition entropy per unit mass is

$$s_i = \frac{k_{\text{B}}n_i}{\rho} \left[1 + \frac{\partial \ln z_i}{\partial \ln T} - (\ln n_i - \ln z_i) \right]. \quad (38)$$

Before pair replacement the partition internal-energy density is

$$u_{\text{part}} = k_{\text{B}}T \sum_i n_i \frac{\partial \ln z_i}{\partial \ln T}. \quad (39)$$

The diagnostic energy components are assembled as

$$u_{\text{trans}} = \frac{3}{2}(n_{\text{tot}} - n_e)k_B T + u_{e^\pm, \text{kin}}, \quad (40)$$

$$u_{\text{rot}} = n_{\text{H}_2} k_B T \frac{\partial \ln Z_{\text{rot}}}{\partial \ln T}, \quad u_{\text{vib}} = n_{\text{H}_2} k_B T \frac{\partial \ln Z_{\text{vib}}}{\partial \ln T}, \quad (41)$$

$$u_{\text{zpe}} = n_{\text{H}_2} E_{\text{zpe}}, \quad u_{\text{diss}} = \frac{1}{2} \chi_{\text{dis}}(n_{\text{H}} + n_{\text{H}^+}), \quad (42)$$

$$u_{\text{ion, H}} = \chi_{\text{ion, H}} n_{\text{H}^+}, \quad u_{\text{ion, He1}} = \chi_{\text{ion, He1}}(n_{\text{He}^+} + n_{\text{He}^{2+}}), \quad (43)$$

$$u_{\text{ion, He2}} = \chi_{\text{ion, He2}} n_{\text{He}^{2+}}. \quad (44)$$

The gas pressure is

$$P_{\text{gas}} = (n_{\text{tot}} - n_e)k_B T + P_{e^\pm}. \quad (45)$$

For the standalone `t13` runtime table, the optional zero-point-energy convention is

$$\epsilon_{\text{runtime}} = \epsilon_{\text{physical}} - X \frac{E_{\text{zpe}}}{2m_{\text{H}}}, \quad (46)$$

while `eps_physical` remains exported. In the union tables this subtraction is disabled so all branches use the same physical energy zero.

5 Relativistic Electron-Positron Pair Gas

The H/He LTE generator includes an ideal relativistic e^-/e^+ gas following the HELM/MESA convention. It enforces

$$n_{e^-} - n_{e^+} = n_{\text{charge}}, \quad (47)$$

where n_{charge} is fixed by the baryonic ion charge. The electron degeneracy parameter η excludes rest mass, and the positron relation uses $-\eta - 2/\theta$ with

$$\theta = \frac{k_B T}{m_e c^2}, \quad \lambda_e = \frac{\hbar}{m_e c}. \quad (48)$$

Using generalized Laguerre nodes y_j and weights w_j with index $\alpha = 1/2$, the code evaluates

$$\gamma_j = 1 + \theta y_j, \quad r_j = \sqrt{2 + \theta y_j}, \quad (49)$$

and stable Fermi factors equivalent to

$$f_j^- = \frac{1}{\exp(y_j - \eta) + 1}, \quad f_j^+ = \frac{1}{\exp(y_j + \eta + 2/\theta) + 1}. \quad (50)$$

The densities are

$$n_{\mp} = \frac{\theta^{3/2}}{\pi^2 \lambda_e^3} \sum_j w_j r_j \gamma_j f_j^{\mp}. \quad (51)$$

The pressure and kinetic-energy densities are

$$P_{\mp} = \frac{m_e c^2 \theta^{5/2}}{3\pi^2 \lambda_e^3} \sum_j w_j y_j r_j^3 f_j^{\mp}, \quad (52)$$

$$u_{\text{kin}, \mp} = \frac{m_e c^2 \theta^{5/2}}{\pi^2 \lambda_e^3} \sum_j w_j y_j r_j \gamma_j f_j^{\mp}. \quad (53)$$

The created-pair rest-mass convention used in the table is

$$u_{\text{rest,pair}} = 2m_e c^2 n_{e^+}. \quad (54)$$

This includes the rest mass of newly created pairs only, not the rest mass of the charge-balancing electrons already tied to the baryonic composition. The lepton entropy density is evaluated from

$$s_{e^\pm} = \frac{u_{e^\pm} + P_{e^\pm} - \eta k_B T n_{\text{charge}}}{T}, \quad (55)$$

and clipped non-negative.

The pair diagnostics written to H/He tables are

$$P_{\text{pair,diag}} = 2P_{e^+}, \quad f_{\text{pair}} = \frac{P_{\text{pair,diag}}}{P_{\text{tot}}}, \quad (56)$$

stored as `ppair` and `pair_pressure_fraction`, plus pair energy, n_{e^-} , n_{e^+} , and η diagnostics. In `t13`, the pair gas replaces the classical ideal electron contribution only for $\log_{10}(T/\text{K}) \geq 7$. In the fully ionized HELM and Coulomb fallback branches, pairs are part of the thermodynamics throughout the branch. In native SCvH and Chabrier dense cells, the imported dense-fluid thermodynamics are not rewritten by pair production; the same HELM-style pair calculation is overlaid only as a diagnostic map.

6 Fully Ionized HELM and Coulomb Fallback Branches

The HELM-style dense fallback assumes fully ionized H/He,

$$n_{\text{H}} = \frac{X\rho}{m_{\text{H}}}, \quad n_{\text{He}} = \frac{Y\rho}{m_{\text{He}}}, \quad n_{\text{charge}} = n_{\text{H}} + 2n_{\text{He}}, \quad (57)$$

and $n_{\text{ions}} = n_{\text{H}} + n_{\text{He}}$. The gas pressure is

$$P_{\text{gas}} = n_{\text{ions}} k_B T + P_{e^\pm}. \quad (58)$$

The specific gas energy is

$$\epsilon_{\text{gas}} = \frac{(3/2)n_{\text{ions}}k_B T + u_{e^\pm}}{\rho} + \epsilon_{\text{offset}}, \quad (59)$$

with an energy-zero offset matched to the fully ionized T13 limit,

$$\epsilon_{\text{offset}} = X \frac{\chi_{\text{dis}}/2 + \chi_{\text{ion,H}}}{m_{\text{H}}} + Y \frac{\chi_{\text{ion,He1}} + \chi_{\text{ion,He2}}}{m_{\text{He}}}. \quad (60)$$

Ion entropies use the ideal translational form

$$s_i = \frac{k_B n_i}{\rho} \left[\frac{5}{2} - (\ln n_i - \ln z_{\text{tr},i}) \right], \quad (61)$$

and pair entropy is added separately. The branch is accepted only where the simple LTE ionization proxy gives

$$x_{\text{H}^+} \geq 0.99, \quad x_{\text{He}^{2+}} \geq 0.90. \quad (62)$$

The SCvH union can also use a Potekhin-Chabrier-like liquid Coulomb corner. With electron-sphere radius

$$a_e = \left(\frac{3}{4\pi n_e} \right)^{1/3}, \quad (63)$$

the ion coupling parameters are

$$\Gamma_{\text{H}} = \frac{e^2}{a_e k_{\text{B}} T}, \quad \Gamma_{\text{He}} = \frac{2^{5/3} e^2}{a_e k_{\text{B}} T}. \quad (64)$$

The reduced liquid free-energy fit implemented in the generator is

$$f_{ii}(\Gamma) = a_1 \left[\sqrt{\Gamma(a_2 + \Gamma)} - a_2 \ln \left(\sqrt{\Gamma/a_2} + \sqrt{1 + \Gamma/a_2} \right) \right] \quad (65)$$

$$+ 2a_3 \left(\sqrt{\Gamma} - \tan^{-1} \sqrt{\Gamma} \right) + b_1 [\Gamma - b_2 \ln(1 + \Gamma/b_2)] + \frac{b_3}{2} \ln(1 + \Gamma^2/b_4), \quad (66)$$

where

$$a_1 = -0.907347, \quad a_2 = 0.62849, \quad a_3 = -\frac{\sqrt{3}}{2} - \frac{a_1}{\sqrt{a_2}}, \quad (67)$$

and

$$b_1 = 0.00456, \quad b_2 = 211.6, \quad b_3 = -10^{-4}, \quad b_4 = 0.00462. \quad (68)$$

The code computes $u_{ii} = \partial f_{ii} / \partial \ln \Gamma$ by a centered finite difference. The Coulomb thermodynamic additions are

$$P_C = \frac{k_{\text{B}} T}{3} \sum_i n_i u_{ii,i}, \quad \epsilon_C = \frac{k_{\text{B}} T}{\rho} \sum_i n_i u_{ii,i}, \quad s_C = \frac{k_{\text{B}}}{\rho} \sum_i n_i (u_{ii,i} - f_{ii,i}). \quad (69)$$

The Coulomb corner is gated by full ionization, $\Gamma_{\text{He}} \leq 175$, and

$$T \geq 3 \max(T_{p,\text{H}}, T_{p,\text{He}}), \quad T_{p,i} = \frac{\hbar}{k_{\text{B}}} \left(\frac{4\pi n_i Z_i^2 e^2}{m_i} \right)^{1/2}. \quad (70)$$

7 SCvH 1995 Union

`scvh_t13_cp_helm_union` combines T13, SCvH, the Coulomb corner, and the HELM fallback. The SCvH helper reads the original hydrogen and helium tables in native $(\log_{10} T, \log_{10} P)$ coordinates, including the hydrogen plasma phase-transition branches. For a target isotherm it builds a composite pressure grid, inserts the low- and high-density PPT branches where applicable, mixes H and He, and then resamples to the AthenaK $\log \rho$ grid with PCHIP.

The H/He volume mixing is the additive-volume law

$$\frac{1}{\rho} = \frac{X}{\rho_{\text{H}}} + \frac{Y}{\rho_{\text{He}}}, \quad (71)$$

and the internal energy is

$$\epsilon_{\text{gas}} = X \epsilon_{\text{H}} + Y \epsilon_{\text{He}}. \quad (72)$$

The entropy is

$$s_{\text{gas}} = X s_{\text{H}} + Y s_{\text{He}} + s_{\text{mix}}, \quad (73)$$

where s_{mix} is the SCvH composition-dependent ideal mixing correction evaluated from the tabulated $\text{H}_2/\text{H}/\text{H}^+$ and $\text{He}/\text{He}^+/\text{He}^{2+}$ fractions and their local derivatives in T and P . The code keeps the full SCvH algebra for this correction rather than replacing it with a composition-independent ideal-gas approximation.

For the gas-only SCvH mixture, the helper computes

$$\alpha = \rho \left[\frac{X}{\rho_{\text{H}}} \left(\frac{\partial \ln \rho_{\text{H}}}{\partial \ln T} \right)_P + \frac{Y}{\rho_{\text{He}}} \left(\frac{\partial \ln \rho_{\text{He}}}{\partial \ln T} \right)_P \right], \quad (74)$$

$$\beta = \rho \left[\frac{X}{\rho_{\text{H}}} \left(\frac{\partial \ln \rho_{\text{H}}}{\partial \ln P} \right)_T + \frac{Y}{\rho_{\text{He}}} \left(\frac{\partial \ln \rho_{\text{He}}}{\partial \ln P} \right)_T \right], \quad (75)$$

then

$$\chi_{\rho, \text{gas}} = \frac{1}{\beta}, \quad \chi_{T, \text{gas}} = -\frac{\alpha}{\beta}. \quad (76)$$

The gas adiabatic gradient follows the mixed entropy derivatives,

$$\nabla_{\text{ad, gas}} = -\frac{(\partial \ln s / \partial \ln P)_T}{(\partial \ln s / \partial \ln T)_P}, \quad (77)$$

and the gas exponents are

$$\Gamma_{1, \text{gas}} = \frac{\chi_{\rho, \text{gas}}}{1 - \chi_{T, \text{gas}} \nabla_{\text{ad, gas}}}, \quad (\Gamma_3 - 1)_{\text{gas}} = \frac{\chi_{\rho, \text{gas}} \nabla_{\text{ad, gas}}}{1 - \chi_{T, \text{gas}} \nabla_{\text{ad, gas}}}. \quad (78)$$

Radiation is then added through the same additive formulas as above. The helper rejects cells whose imported thermodynamic derivatives disagree too strongly with finite differences on the resampled table.

The union ownership rule is

1. use valid SCvH cells wherever available,
2. otherwise use the fully ionized Coulomb corner where its gates pass,
3. otherwise use the HELM fully ionized pair fallback in hot dense cells,
4. otherwise use T13 on the low-density side,
5. fill remaining rectangular holes from T13 or HELM and enforce a strictly increasing $\ln \epsilon$ envelope along each temperature row.

The source-map fields are `source_t13`, `source_scvh`, `source_corner_cp`, `source_corner_helm`, `source_corner`, `source_filler`, and `eps_envelope_correction`.

8 Chabrier 2021 Dense-Fluid Union

`chabrier2021_t13_helm_union` is the current default dense H/He EOS product. It covers

$$-20 \leq \log_{10} \rho_{\text{cgs}} \leq 3, \quad 0 \leq \log_{10}(T/\text{K}) \leq 10 \quad (79)$$

in the hardcoded 640 by 640 job. The dense-fluid input tables are available at helium mass fractions $Y = 0.275, 0.292, 0.297$; the helper linearly interpolates or, for the local default $Y = 0.30$, mildly extrapolates in Y .

The native Chabrier rows are in $(\log_{10} T, \log_{10} P_{\text{GPa}})$. They contain $\log_{10} \rho$, $\log_{10} u$, $\log_{10} s$, the two density derivatives at constant P and T , and ∇_{ad} . Unit conversion is

$$P_{\text{gas}} = 10^{\log_{10} P_{\text{GPa}} + 10}, \quad \epsilon_{\text{gas}} = 10^{\log_{10} u_{\text{MJ kg}^{-1}} + 10}, \quad s_{\text{gas}} = 10^{\log_{10} s_{\text{MJ kg}^{-1} \text{K}^{-1}} + 10}. \quad (80)$$

For each target isotherm, valid native cells are filtered, forced monotonic in $\log \rho$, and linearly interpolated to the target $\log \rho$ grid.

The gas derivatives are

$$\chi_{\rho,\text{gas}} = \left[\left(\frac{\partial \ln \rho}{\partial \ln P} \right)_T \right]^{-1}, \quad \chi_{T,\text{gas}} = - \frac{(\partial \ln \rho / \partial \ln T)_P}{(\partial \ln \rho / \partial \ln P)_T}, \quad (81)$$

and

$$\delta_{\text{gas}} = - \left(\frac{\partial \ln \rho}{\partial \ln T} \right)_P. \quad (82)$$

The heat capacities are reconstructed as

$$c_{p,\text{gas}} = \frac{P_{\text{gas}} \delta_{\text{gas}}}{\rho T \nabla_{\text{ad,gas}}}, \quad c_{v,\text{gas}} = c_{p,\text{gas}} - \frac{P_{\text{gas}} \chi_{T,\text{gas}}^2}{\rho T \chi_{\rho,\text{gas}}}. \quad (83)$$

After adding radiation, the closure is

$$\chi_{\rho} = \frac{P_{\text{gas}} \chi_{\rho,\text{gas}}}{P_{\text{tot}}}, \quad \chi_T = \frac{P_{\text{gas}} \chi_{T,\text{gas}} + 4P_{\text{rad}}}{P_{\text{tot}}}, \quad (84)$$

$$c_v = c_{v,\text{gas}} + \frac{4a_{\text{rad}} T^3}{\rho}, \quad \Gamma_3 - 1 = \frac{P_{\text{tot}} \chi_T}{\rho T c_v}, \quad (85)$$

$$\Gamma_1 = \chi_{\rho} + \chi_T (\Gamma_3 - 1), \quad c_s^2 = \Gamma_1 \frac{P_{\text{tot}}}{\rho}. \quad (86)$$

The union rule is:

1. build T13 everywhere as the low-density chemical EOS,
2. import and resample the Chabrier dense-fluid EOS,
3. fill tiny dense Chabrier holes row-by-row with monotone PCHIP or nearest valid edge values,
4. overlay HELM pair diagnostics on valid Chabrier cells without changing their pressure, energy, or adiabatic indices,
5. blend T13 and Chabrier over $10^{-3} \leq \rho_{\text{cgs}} < 10^{-2}$ with smotherstep weights,
6. additionally blend the first few dense Chabrier onset cells in temperature to reduce a visible Γ_1 boundary,
7. use pure Chabrier cells at higher density wherever valid,
8. use the HELM fully ionized fallback above $\log_{10}(T/\text{K}) \geq 7$ where Chabrier is not valid.

The smotherstep weight is

$$S(q) = 6q^5 - 15q^4 + 10q^3, \quad 0 < q < 1, \quad (87)$$

with $S = 0$ below the interval and $S = 1$ above it. Linear physical fields are blended as $(1-S)A + SB$; logarithmic runtime fields are then rebuilt from the blended pressure, energy, and sound speed. On pure Chabrier dense cells, species fractions and μ are diagnostic proxies from the simple atomic H/He Saha solver and do not feed back into the Chabrier thermodynamics.

9 Validation and Generated Artifacts

Before writing, each table must have finite positive runtime fields, positive c_s^2 , positive Γ_1 and $\Gamma_3 - 1$, positive μ , bounded species fractions, bounded β_{rad} , and non-negative pair-pressure diagnostics. Union tables additionally validate binary non-overlapping source masks, finite pair diagnostics, and strictly increasing $\ln \epsilon$ along every temperature row after any rectangular filling.

The maintained generated tables live under `eos_tables/`. The Chabrier 640 by 640 diagnostic table can exceed GitHub’s normal 100 MB object limit and should be handled through Git LFS or an external artifact store if it needs to be versioned.

The maintained summary figures under `eos_validation_test/plots/` are:

- `saha_hydrogen_summary.png`,
- `lte_t13_prad_eos_summary.png`,
- `scvh_t13_cp_helm_union_prad_640_summary.png`,
- `chabrier2021_t13_helm_union_prad_640_summary.png`.

These are one current summary per maintained EOS table. The older pair/no-pair comparison PNGs were diagnostic snapshots and are not part of the maintained summary set.

10 Source Literature Encoded by the Implementation

The H/He low-density chemistry follows the Tomida et al. Appendix-1 partition-function EOS as used by the local Andelman-style workflow. Dense low-temperature H/He fluid physics comes either from Saumon, Chabrier, and Van Horn 1995 or from the Chabrier et al. 2021 H/He tables. The fully ionized fallback pair thermodynamics follow the ideal relativistic HELM/MESA electron-positron gas convention associated with Timmes and Swesty/Timmes and Arnett. The optional Coulomb corner uses a Potekhin-Chabrier-style liquid free-energy fit for fully ionized ions. The table metadata records the exact source URLs and scalar constants used for a generated table.